# A3-PS015-TC001

## Explainable Home and Last-Mile Connectivity Diagnosis


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**AIORI Hackathon 3**

October 14, 2026

# Problem Description

- Standard peak tests overlook loaded jitter, packet loss, and round-trip latency, critical for real-time application quality.
- Performance fluctuates unpredictably across underlying media (fiber, cellular, fixed wireless, satellite).

- Traditional tools fail to explain findings with evidence or generate clean diagnostic escalation reports.

# Background

The Internet Standard framework for IP performance metrics (RFC 2330) provides rigorous guidelines for measuring connectivity reliability across packet networks.

By coordinating scheduled probes across endpoints, we establish baseline metrics for latency, DNS resolution, and throughput while isolating local Wi-Fi degradation.

### Core Measurement Metrics

- ✓ Gateway Reachability & Response
- ✓ DNS Resolution Latency
- ✓ Induced Packet Loss & Jitter

# Literature & RFC Review

### RFC 2330 Framework

Foundational criteria for active and passive IP performance metrics measurement across host endpoints.

### ITU-T Y.1541

Network performance objectives for IP-based services, establishing QoS classes for latency and jitter.

### Active Probe Studies

Prior work on scheduled lightweight telemetry to distinguish local congestion from ISP backhaul bottlenecks.

# Proposed Solution

## AIORI 3 Proposed Solution

Detailed description of the overall solution:

We propose a lightweight desktop diagnostic agent that coordinates active network probes inside a controlled lab environment. The system features an explainable decision-rule engine that maps measured reachability, DNS latency, and packet loss to transparent fault categories, returning "Inconclusive" with an evidence summary when fault isolation is ambiguous.

# Optimization Proposed

## AIORI 3 Solution Optimization

List of optimizations and enhancements compared to traditional tests:

| Controlled Fault | Traditional Speed Test | Proposed Explainable Agent |
|---|---|---|
| **Gateway Unreachable** | Generic Connection Failed | Local ARP & Gateway ICMP check |
| **DNS Failure** | Browser Timeout | Targeted DNS resolution probe |
| **Induced Loss** | Slow download speed | Packet drop % & jitter analysis |
| **Network Congestion** | Lower Mbps rating | Loaded vs unloaded RTT delta |

# Solution Architecture & Design

## AIORI 3 Architecture Design (UML Diagram)



**[System Architecture UML Diagram]**

Desktop Agent $\rightarrow$ Active Probe Module $\rightarrow$ Decision Engine $\rightarrow$ Escalation Report

# Data Flow Diagram

## AIORI 3 Data Flow Diagram



**[Data Flow Diagram (DFD)]**

Probes $\rightarrow$ Raw Metrics Collection $\rightarrow$ Rule Matching $\rightarrow$ Diagnosis / Inconclusive Case

# Deployment Timeline

## AIORI 3 Deployment Timeline

### Phase 1: Setup

Private GitHub repo initialization and setting up the controlled lab testbed.

### Phase 2: Injection

Scripting 4 controlled fault injections and implementing rule matching.

### Phase 3: Validation

Inconclusive scenario analysis, demo preparation, and report generation.

# Bibliography

## AIORI 3 Bibliography

**[1]** V. Paxson, G. Almes, J. Mahdavi, M. Mathis. "Framework for IP Performance Metrics", RFC 2330, May 1998.

**[2]** ITU-T Recommendation Y.1541. "Overall network performance objectives for IP-based services", International Telecommunication Union, 2011.